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- a. List the ID and description of all items never used in any appointment service.
[1 mark]

RESULT = (($\pi_{\text{item_id}}$ ITEM) - ($\pi_{\text{item_id}}$ APPTSERVICE_ITEM)) $\bowtie$ ($\pi_{\text{item_id}, \text{item_desc}}$ ITEM)

- b. List the patient number, first name, last name, emergency contact first name, emergency contact last name and emergency contact phone number of all patients who live in a city named Mooroolbark and had an appointment on 08 September 2024.
[2 marks]

MOOROOLBARK_PATIENTS = $\pi_{\text{patient_no}}$ ($\sigma_{\text{patient_city} = \text{'Mooroolbark'}}$ PATIENT)

EIGHT_SEPT_PATIENTS = $\sigma_{\text{appt_datetime} = \text{'8/09/2024'}}$ APPOINTMENT

CONTACTS = $\pi_{\text{ec_id}}$ EMERGENCY_CONTACT

RESULT = $\pi_{\text{patient_no}, \text{patient_fname}, \text{patient_lname}, \text{ec_fname}, \text{ec_lname}, \text{ec_phone}}$ (MOOROOLBARK_PATIENTS $\bowtie$ EIGHT_SEPT_PATIENTS $\bowtie$ CONTACTS)

- c. List the number, first name, last name and email address of all patients attended by endodontists (i.e. providers who specialise in ENDODONTICS).
[3 marks]

ENDODONTISTS = $\pi_{\text{spec_id}}$ ($\sigma_{\text{spec_name} = \text{'ENDODONTICS'}}$ SPECIALISATION)

PROVIDER_CODES = $\pi_{\text{provider_code}, \text{spec_id}}$ PROVIDER $\bowtie$ ENDODONTISTS

ENDO_PATIENTS = $\pi_{\text{patient_no}, \text{provider_code}}$ APPOINTMENT $\bowtie$ PROVIDER_CODES

RESULT = $\pi_{\text{patient_no}, \text{patient_fname}, \text{patient_lname}, \text{patient_contactemail}}$ $\bowtie$ ENDO_PATIENTS